$$\int \frac{dy}{dx} = \int 2x\, dx$$

$$= x^2 + C$$

y(x) is the solution

$$\boxed{y = x^2 + C}$$

$$\int \overset{h(y)}{dy} = \int \overset{g(x)}{2x\, dx}$$

$$\vec{F} = m\vec{a}$$

$$\vec{F} = \frac{d\vec{p}}{dt}$$, where $\vec{p}$ is the momentum

$$\frac{dP}{dt} = rP \sim \frac{dP}{dt} \propto P$$

$$\int \frac{1}{P}\, dP = \int r\, dt$$

$$\ln|P| = rt + C$$

$$\rightarrow e^{\ln|P|} = e^{rt+C}$$

$$\rightarrow P(t) = e^{rt+C}$$

$$= e^{rt} e^{C}$$

$$\boxed{P(t) = Ke^{rt}}$$

$$\boxed{\begin{aligned} e^{\ln(x)} &= x \\ \ln(e^x) &= x \end{aligned}}$$

$$\boxed{x^{a+b} = x^a x^b}$$

$$\frac{dN}{dt} = -\lambda N$$

$$\int \frac{1}{N}\, dN = \int -\lambda\, dt$$

$$\ln N = -\lambda t + C$$

$$N(t) = e^{-\lambda t + C}$$

$$= e^C e^{-\lambda t}$$

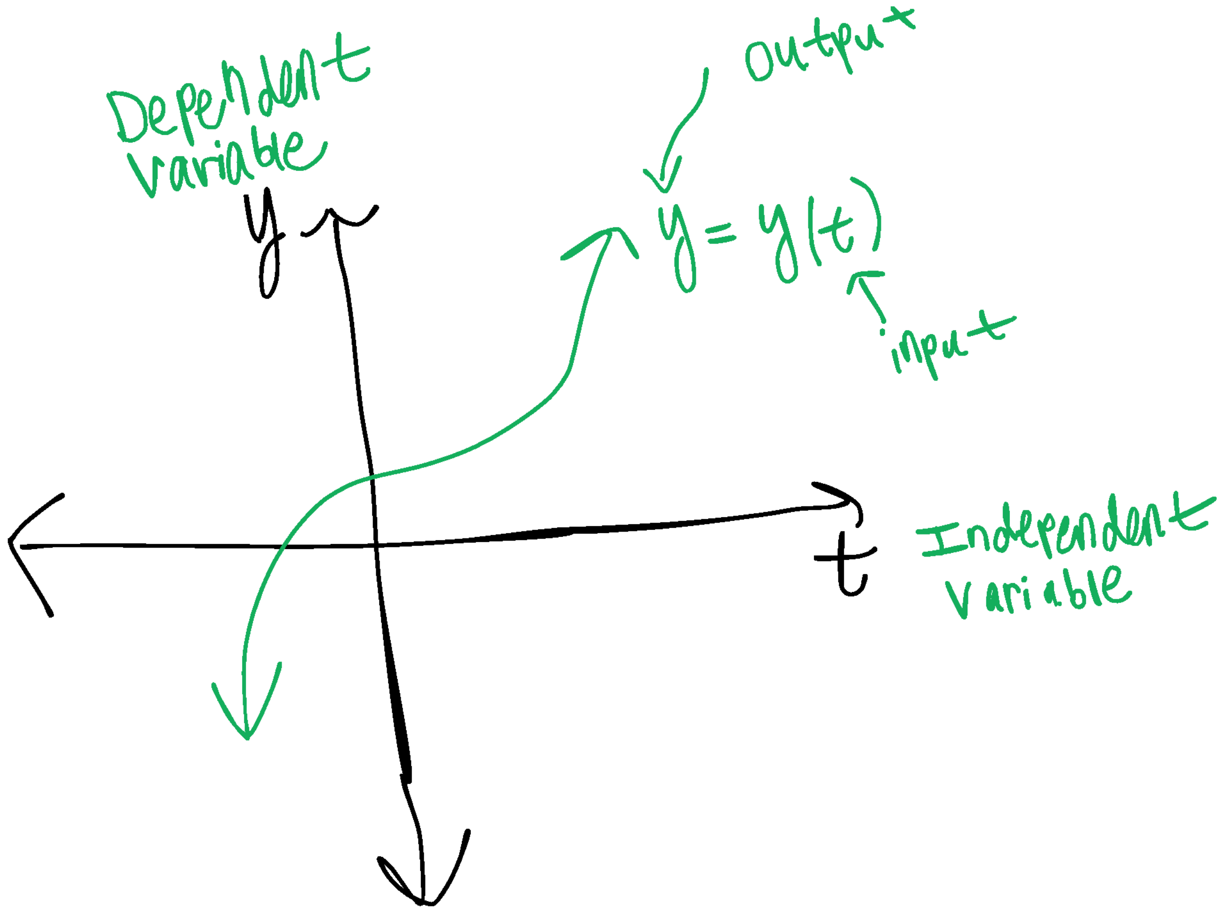
$$N(t) = \frac{K}{e^{\lambda t}}$$

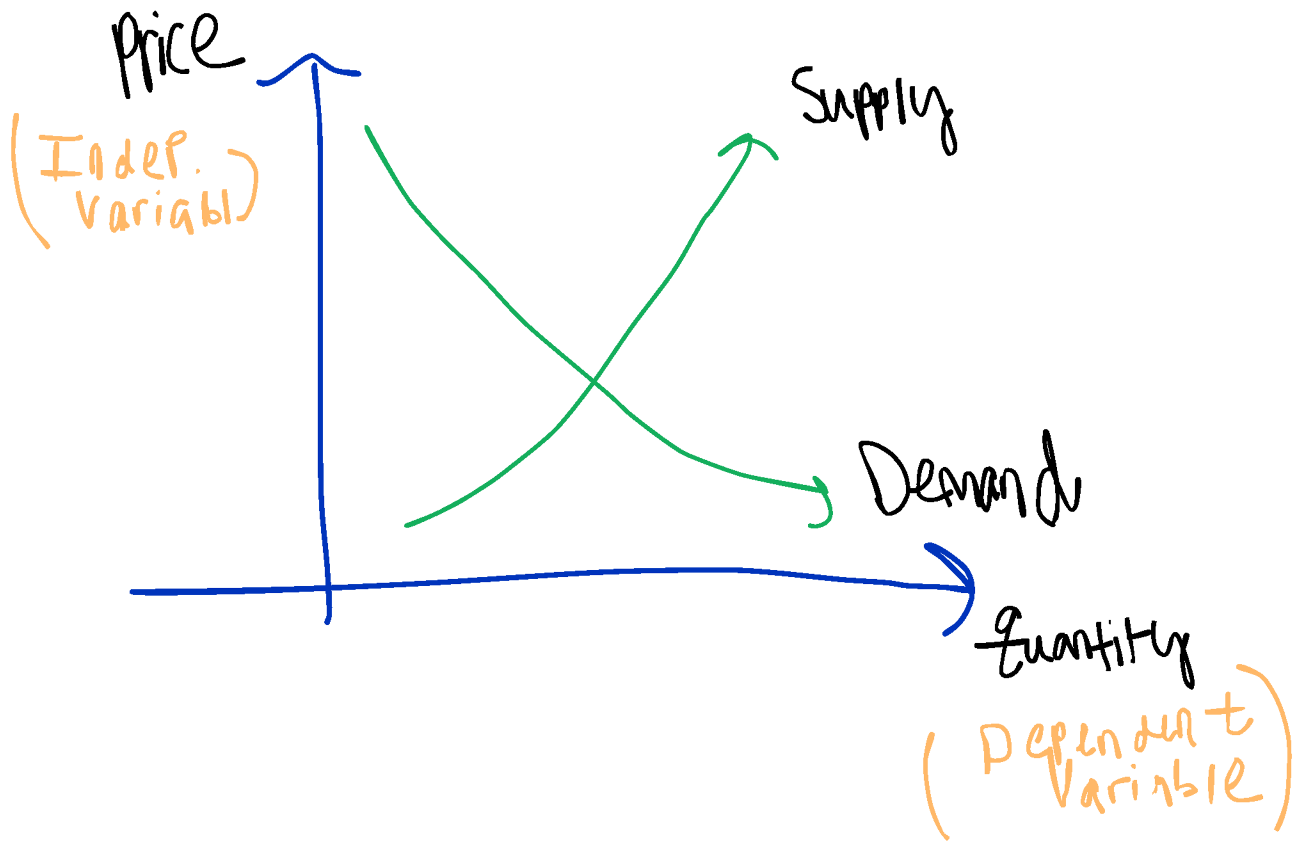
$$\lim_{t\to\infty} N(t) = \lim_{t\to\infty} \frac{K}{e^{\lambda t}} = 0$$

$$\frac{dy}{dt} = \lambda t$$

$\lambda > 0$ Exp. growth

$\lambda < 0$ Exp. decay


Dependent Variable
y
output
y = y(t)
input
t
Independent Variable


Price
(Indep. Variabl)
Supply
Demand
Quantity
(Dependent Variable)

$$f(x,y) = xy^2 + x^2y + x^2 + y^2$$

$$\frac{\partial f}{\partial x}\ (y\text{-const}) = y^2 + 2xy + 2x$$

$$\frac{\partial f}{\partial y} = 2yx + x^2 + 2y$$

PDE

Partial DE

$$\frac{\partial f}{\partial x} = x^2 + 2xy + 2x$$

The power of the highest derivative is 3, the degree of the ODE

order 2

$$\frac{d^2y}{dx^2} = y''$$

$$\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2}$$

$$(y'')^3 + (y')^5 + y = 0$$

2nd derivative is the highest derive

$$f(x) = 3x^5 + 8x^4 + 2x + 3$$

$$\deg(f(x)) = 5$$

$$g(x) = e^{x^5} + (x^5)^{e^x}$$

$$\deg(g(x))\ \text{DNE}$$

Linearity

This concept will recur throughout the course; we must define it with care.

A

n ODE in the dependent variable y and independent variable x is called **linear** if it can be written in the form

$$a_n(x)\frac{d^ny}{dx^n} + a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_1(x)\frac{dy}{dx} + a_0(x)\,y \;=\; g(x),$$

where the coefficients $a_0(x), a_1(x), \ldots, a_n(x)$ and the forcing term $g(x)$ depend at most on the *independent* variable x. An ODE that cannot be cast in this form is called **nonlinear**.

The defining features of linearity are:

(i) The dependent variable y and each of its derivatives appear to the **first power only**.

(ii) There are **no products** of y with its derivatives (no $y \cdot y'$, no $(y')^2$, etc.).

(iii) No **transcendental functions** (sine, cosine, exponential, logarithm, etc.) are applied to y or its derivatives.

(iv) Coefficients may be arbitrary functions of x alone.

e.g. Linear ODE

$$\underbrace{3x^2}_{a_3(x)}\frac{d^3y}{dx^3} + \underbrace{e^x}_{a_2(x)}\frac{d^2y}{dx^2} + \underbrace{2}_{a_1(x)}\frac{dy}{dx} + \underbrace{8}_{-g(x)} = 0$$

order = 3

degree = 1

$v(t) = 16t^2 + 4t + 3$

General Solution $\rightarrow$ $s(t) = \int v(t)\,dt = \frac{16}{3}t^3 + 2t^2 + 3t + C$

Say $s(0) = 0$

So $0 = 0 + 0 + 0 + C$

$\rightarrow C = 0$

Particular Solution

So $s(t) = \frac{16}{3}t^3 + 2t^2 + 3t$

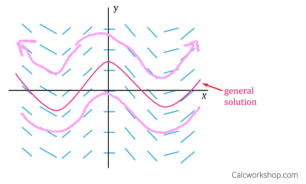
T

he **general solution** of an nth-order ODE is a family of functions containing n arbitrary constants $C_1, C_2, \ldots, C_n$, encompassing all possible solutions of the equation. **A particular solution** is any specific member of this family, obtained by assigning numerical values to the constants. An **Initial Value Problem** (IVP) consists of an ODE together with sufficient initial conditions to determine these constants uniquely.

The Initial Value Problem

A first-order IVP has the form

$$\frac{dy}{dx} = f(x, y), \qquad y(x_0) = y_0.$$



Module 1 Exercises. Work each problem carefully on a separate sheet; full justification is expected.

(A) Classification (state order, degree, and linearity for each).

1.1 $\left(\frac{d^3y}{dx^3}\right)^2 + 5\left(\frac{dy}{dx}\right)^4 + 7y = \cos(x)$. 3, 2, not

1.2 $\frac{d^2y}{dx^2} + e^x\frac{dy}{dx} + y\sin(x) = x^3$. 2, 1, linear

1.3 $\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \frac{d^2y}{dx^2}$. *(Hint: rationalize first.)* 2, ~~2~~, 1, not

$$\left(\frac{d^2y}{dx^2}\right)^2 = 1 + \left(\frac{dy}{dx}\right)^2 \leftrightarrow$$

$$\rightarrow \left(\frac{d^2y}{dx^2}\right)^2 - \left(\frac{dy}{dx}\right)^2 = 1$$

(B) Verification.

1.4 Verify by direct substitution that $y(x) = e^{-2x}\sin(3x)$ is a solution of

$$y'' + 4y' + 13y = 0.$$

1.5 Verify that the implicit relation $x^2 + y^2 = C$ defines a family of solutions to the ODE

$$\frac{dy}{dx} = -\frac{x}{y}.$$

$$\sqrt{x+y} \neq \sqrt{x} + \sqrt{y}$$

$$\sqrt{0} = \sqrt{1 + (-1)} \qquad \sqrt{1} + \sqrt{-1}$$

$$0 = \qquad \neq \qquad 1 + i$$

$$y = \left(e^{-2x}\right)\left(\sin(3x)\right)$$

$$y' = \left(-2e^{-2x}\right)\left(\sin(3x)\right) + \left(e^{-2x}\right)\left(3\cos(3x)\right)$$

$$= e^{-2x}\left[-2\sin(3x) + 3\cos(3x)\right]$$

$$y'' = \left(-2e^{-2x}\right)\left(-2\sin(3x) + 3\cos(3x)\right) + \left(e^{-2x}\right)\left(6\cos(3x) - 9\sin(3x)\right)$$

$$= e^{-2x}\left[4\sin(3x) - 6\cos(3x) + 6\cos(3x) - 9\sin(3x)\right)$$

$\frac{d}{dx}x^n = nx^{n-1}$

$(uv)' = u'v + uv'$

$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$

$(f(g(x)))' = f'(g(x))g'(x)$

$\frac{d}{dx}e^x = e^x$

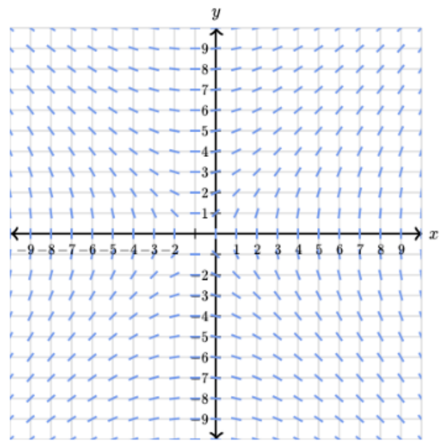
$\frac{d}{dx}e^{g(x)} = e^{g(x)} \cdot g'(x)$

$\frac{d}{dx}e^{3x} = 3e^{3x}$

$\frac{d}{dx}\sin x = \cos x$

$\frac{d}{dx}\sin(g(x)) = \cos(g(x)) \cdot g'(x)$

$\frac{d}{dx}\sin(3x) = 3\cos(3x)$



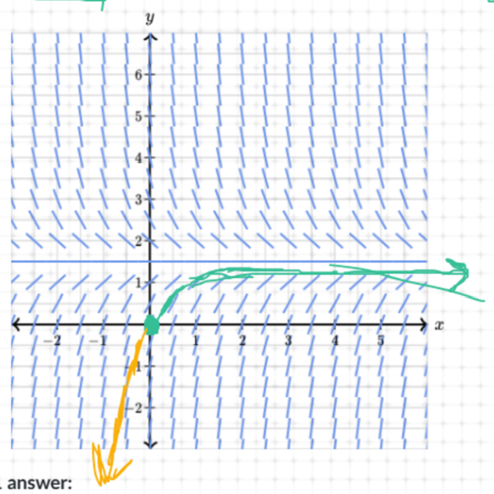
Choose 1 answer:

- (A) $\dfrac{dy}{dx} = \dfrac{x}{y}$
- (B) $\dfrac{dy}{dx} = \dfrac{x}{y-1}$
- (C) $\dfrac{dy}{dx} = \dfrac{x}{y+1}$
- (D) $\dfrac{dy}{dx} = \dfrac{x-1}{y}$
- (E) $\dfrac{dy}{dx} = \dfrac{x+1}{y}$

$y'(-1) = 0$ $\dfrac{dy}{dx} = 0$ when $x = -1$

If the initial condition is $(0, 0)$, what is the range of the solution curve $y = f(x)$ for $x \geq 0$?

set of all y-value



$\lim\limits_{x \to \infty} y = 1.5$

Choose 1 answer:

- (A) $(-\infty, 1.5)$
- (B) $(0, 1)$
- (C) $[0, 1.5)$

Range $[0, 1.5)$